MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 3ACP1A, 3ACP2A, AND 3ACP7A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter: Unit:	Ef V	Ec1 Vdc	Eb1 Vdc	Eb2 Vdc (Note 1)	Eb3 Vdc	ed v	Rg Meg	Zd Meg (Note 2)	Ehk Vdc	Eb3/Eb2 Ratio (Note 3)	Barometric pressure, reduced mmHg
Maximum:	6.9	-200	1,500	2,200	6,600	550	1.5	1.0	±180	3.0	87.0
Minimum:	5.7	0	---	1,000	1,000	---	---	---	---	1.0	---
TEST CONDITIONS:	6.3	Adjust	Focus	2,000	4,000	---	---	---	---	2.0	---

GENERAL:

Qualification - Required

(J) denotes changes

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METHOD	REQUIREMENT OR TEST	TYPE	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
(J)	Qualification inspection							
1141	Pressure (implosion)	All	-		---	---	---	---
1002	Barometric pressure, reduced	All	-	87.0 mmHg	---	---	---	---
5111	Vibration	All	-		Width	---	1.0	mm
1331	Direct-interelectrode capacitance	All	-	k to all	Ck	---	4.2	pF
				g1 to all	Cg1	---	5.5	pF
				D1 to D2	C1D2	---	2.1	pF
				D3 to D4	C3D4	---	1.5	pF
				D1 to all	CD1	---	5.8	pF
				D2 to all	CD2	---	5.8	pF
				D3 to all	CD3	---	4.5	pF
				D4 to all	CD4	---	4.5	pF
5201	Electrode current (anode No. 3)	All	-	Ec1 = 0	Ib	500	---	μAdc
5101	Neck and bulb alinement (electrostatic types)	All	-		---	---	2.25	inch
5216	Cathode illumination	All	-		---	---	---	---
5216	Stray light emission (conventional types)	All	-	Eb2 = 2,200 Vdc; Eb3 = 6,600 Vdc	---	---	---	---
5248	Deflection factor	All	-	1D2; Eb3 = Eb2 = 2,000 Vdc	DF	142	163	Vdc/in.
				3D4; Eb3 = Eb2 = 2,000 Vdc	DF	110	132	Vdc/in.
5248	Deflection-factor uniformity	All	-		---	---	2.0	%
5248	Deflection-factor uniformity	All	-	Eb3 = Eb2 = 2,000 Vdc	---	---	3.0	%
5115	Shock	All	-		---	---	---	---
1216	Base material insulating quality	All	-		---	---	---	---
	Quality conformance inspection, part 1							
5201	Voltage breakdown	All	-		---	---	---	---
5206	Gas "cross"	P1A	4	Ib3 = 50 μAdc	---	---	---	---
		P2A	4	Ib3 = 200 μAdc	---	---	---	---
		P7A	4	Ib3 = 200 μAdc	---	---	---	---
5106	Screen and faceplate blemishes	All	-		---	---	---	---
5223	Modulation	All	-	Ib3 = 25 μAdc	ΔEc1	---	21	Vdc
				Ib3 = 200 μAdc	ΔEc1	---	45	Vdc
5231	Spot position (electrostatic deflection)	All	-		---	---	10	mm
5231	Spot displacement (leakage)	All	-		---	---	5	mm
5241	Grid cutoff voltage	All	-		Ec1	-75	-45	Vdc
5103	Pattern distortion	All	-		---	---	2	%
5251	Grid No. 1 leakage current	All	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1 -Continued							
5251	Anode No. 1 leakage current	All	-		---	---	3	µAdc
5251	Anode No. 2 leakage current	All	-		---	---	---	---
5221	Light output	P1A	-	Ib3 = 100 µAdc	Light	80	---	FtL
	Quality conformance inspection, part 2							
1301	Heater current	All	-		If	540	660	mA
5201	Electrode current (anode No. 1)	All	-	Ib3 = 500 µAdc	Ib1	-15	+5	µAdc
5201	Electrode current (cathode)	P1A P2A P7A	- - -	Ib3 = 50 µAdc Ib3 = 200 µAdc Ib3 = 200 µAdc	Ik Ik Ik	--- --- ---	125 600 600	µAdc µAdc µAdc
5101	Base alinement (electrostatic types)	All	-	1D2, pin No. 5	---	---	---	---
5101	Side terminal alinement (electrostatic types)	All	-	1D2	---	---	---	---
5101	Side terminal and base alinement	All	-	Pin No. 5	---	---	---	---
5101	Neck and base alinement (electrostatic types)	All	-		---	---	---	---
(J) 1111	Base pin solder depth	All	-		---	---	---	---
5221	Screens	All	-		---	---	---	---
5226	Line width "A" (electrostatic deflection)	All	-	Ib3 = 50 µAdc Ib3 = 200 µAdc	Width Width	--- ---	0.4 0.7	mm mm
5101	Angle between traces	All	-		---	89	91	Degrees
5226	Line width "B" (electrostatic deflection)	All	-	Ib3 = 50 µAdc Ib3 = 200 µAdc	Width Width	--- ---	0.6 0.8	mm mm
5246	Focusing voltage at cutoff	All	-		Eb1	390	---	Vdc
5246	Focusing voltage (zero-bias)	All	-		Eb1	---	550	Vdc
5248	Deflection factor	All	-	1D2	DF	175	200	Vdc/in.
5248	Deflection factor	All	-	3D4	DF	138	158	Vdc/in.
5251	Heater-cathode leakage current	All	-		Ihk	---	15	µAdc
1101	Secureness of base, cap, or insert	All	-		---	---	---	---
(J) 1105	Permanence of marking	All	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPE	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3							
---	Life test	All	-	Group C; Ib3 = 30 µAdc; t = 500 hours (min)	---	---	---	---
---	Life-test end points:							
5223	Modulation	All	-	Ib3 = 150 µAdc	ΔEc1	---	45	Vdc
5226	Line width "A"	All	-	Ib3 = 150 µAdc	Width	---	0.8	mm
5226	Line width "B"	All	-	Ib3 = 150 µAdc	Width	---	0.9	mm

NOTES:

1. Accelerator power input (average) should be limited to 6 watts.
2. The deflection electrode circuit resistances should be approximately equal. Higher resistance values up to 5.0 Megohms may be used for low-beam current operation.
3. This tube is designed for optimum performance when operating at an Eb3/Eb2 ratio of 2.0. Operation at other ratios of Eb3/Eb2 may result in changes in deflection uniformity and pattern distortion.
4. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

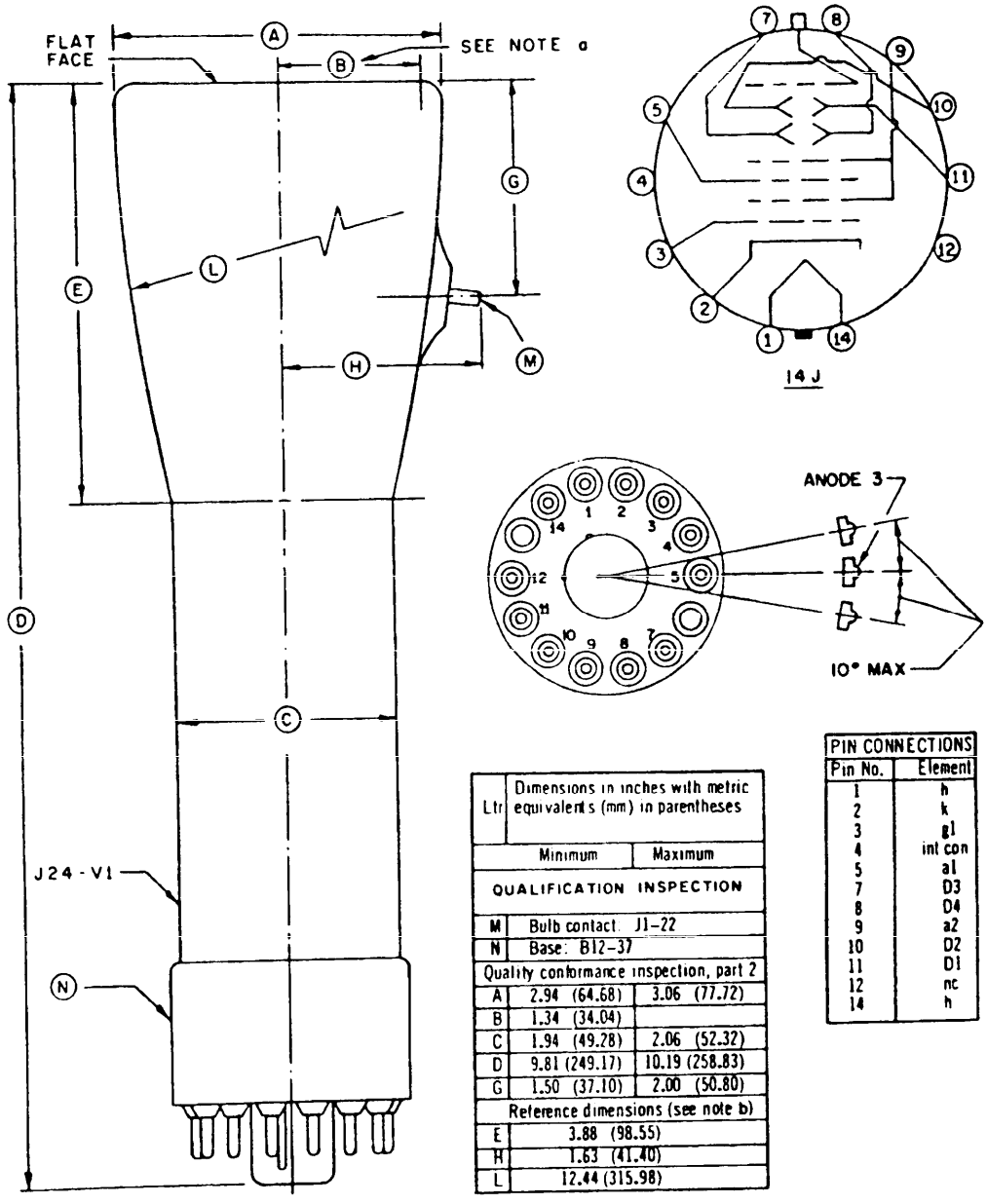
User activities:
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Preparing activity:
Navy - EC

Agent:
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
QUALIFICATION INSPECTION		
M	Bulb contact: J1-22	
N	Base: B12-37	
Quality conformance inspection, part 2		
A	2.94 (64.68)	3.06 (77.72)
B	1.34 (34.04)	
C	1.94 (49.28)	2.06 (52.32)
D	9.81 (249.17)	10.19 (258.83)
G	1.50 (37.10)	2.00 (50.80)
Reference dimensions (see note b)		
E	3.88 (98.55)	
H	1.63 (41.40)	
L	12.44 (315.98)	

PIN CONNECTIONS	
Pin No.	Element
1	h
2	k
3	g1
4	int con
5	a1
7	D3
8	D4
9	a2
10	D2
11	D1
12	nc
14	h

NOTES:
a. Minimum useful screen radius.
b. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 3ACP1A, 3ACP2A, and 3ACP7A.

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